

LESSON 118 (2.0 GROUND)**Lesson Objective**

The instructor will review the regulations applicable to a student pilot preparing for solo flight including both 14 CFR part 61 and 91. Aircraft systems and their possible malfunctions, TOLD cards, and performance charts will be reviewed as well.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Aeronautical Information Manual. ([AIM](#))
 - Chapter 03 - Airspace.
 - Section 02 - Radio Communications Phraseology and Techniques.
 - Chapter 06 - Emergency Procedures.
 - Chapter 07 - Safety of Flight.
 - Section 04 - Wake Turbulence.
 - Section 06 - Potential Flight Hazards.
- Federal Aviation Regulations.
- FIT FAA Training Flight Operations Manual.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 05 - Aerodynamics of Flight.
 - Chapter 07 - Aircraft Systems.
 - Chapter 09 - Flight Manuals and Other Documents.
 - Chapter 15 - Airspace.
- Warrior III Pilot's Information Manual.
 - Section 04 - Normal Procedures.
 - Section 07 - Description and Operations of the Airplane and its Systems.

Lesson Content

This lesson serves as a review before the stage check mock. Quiz the student on all of the line items below.

V Speeds and Emergencies Quiz

- Administer the blank V speeds and emergencies quiz.

System and Equipment Malfunctions

- Give the student different malfunction scenarios.
 - Loss of oil pressure, high oil temperature, carburetor icing, hole in the exhaust shroud, etc.

Regulations Applicable to Student Pilot:

- Part 61
 - Required documents - 61.3:
 - Pilot certificate, medical, and government photo ID.
 - Drugs and alcohol - 61.15 and 61.16:
 - Outline the consequences.
 - Student pilot certificate duration - 61.19:
 - No expiration date if issued after April 1st, 2016.
 - Medical certificate classes and durations - 61.23:
 - Special issuance, SODA, etc. (67.401)
 - No need to get into BasicMed.
 - Pilot Logbooks - 61.51:
 - Information to be logged.
 - Medical Deficiency - 61.53:
 - What constitutes a medical deficiency?
 - Recent flight experience - 61.57:
 - Acting as PIC while carrying passengers during daytime vs nighttime.
 - Same category, class, and type.
 - Must be to a full stop if you are in a tailwheel airplane.
 - Change of address - 61.60:
 - FAA must be notified in writing within 30 days.
 - Solo requirements - 61.87:
 - This will be covered under endorsements required for solo flight below.
 - General limitations (SP) - 61.89:
 - A student pilot may not act as pilot in command of an aircraft:
 - That is carrying a passenger.
 - That is carrying property for compensation or hire.
 - For compensation or hire.
 - In furtherance of a business.
 - On an international flight, except that a student pilot may make solo training flights from Haines, Gustavus, or Juneau, Alaska, to White Horse, Yukon, Canada, and return over the province of British Columbia.
 - With a flight or surface visibility of less than 3 statute miles during daylight hours or 5 statute miles at night.
 - When the flight cannot be made with visual reference to the surface; or
 - In a manner contrary to any limitations placed in the pilot's logbook by an authorized instructor.
 - Solo operations in class b airspace - 61.95:
 - This will be covered under endorsements required for solo flight below.

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- Part 91
 - Pilot in Command (PIC) responsibilities - 91.3:
 - PIC is responsible for airworthiness, safety of flight, and has the final authority.
 - Aircraft airworthiness - 91.7:
 - Cannot fly unless airworthy.
 - Flight manual, placards, and markings - 91.9:
 - All must be on the airplane to be airworthy.
 - Careless and reckless operation - 91.13:
 - May not operate in a manner that endangers others, on the ground or in the air.
 - Drugs and alcohol - 91.17:
 - Cannot be under the influence.
 - 8 hours “bottle to throttle” and a maximum BAC of 0.04%.
 - Compare to FIT’s regulation.
 - Must submit to a drug/alcohol test if requested.
 - Must furnished results to the FAA within 4 hours.
 - Preflight action - 91.103:
 - IFR flight or away from the vicinity of an airport.
 - NOTAMs.
 - Weather.
 - Known delays.
 - Runway lengths.
 - Alternates if needed.
 - Fuel requirements.
 - Takeoff/landing data.
 - Crewmembers at stations - 91.105:
 - When should seatbelts and shoulder harnesses be worn?
 - Use of safety belts and harnesses - 91.107:
 - All occupants must be briefed on how to use seatbelts and shoulder harnesses.
 - When should seatbelts and shoulder harnesses be worn?
 - Operating near other aircraft - 91.111:
 - May not operate an aircraft so close to another that it creates a collision hazard.
 - Review formation flight.
 - Right of way - 91.113:
 - See and avoid other aircraft if the weather permits!
 - Review all of the different scenarios and who has the right of way.
 - Minimum safe altitudes - 91.119:
 - Congested → 1,000 ft above, 2,000 ft horizontally.
 - Other than congested → 500 ft above.
 - Sparsely populated and open water → cannot operate 500' away from the nearest vessel, structure, person, or vehicle.

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- Compliance with ATC - 91.123:
 - Must obey instructions unless amended clearance or emergency.
 - Must notify ATC immediately when deviating.
 - May need to submit a report to ATC within 48 hours.
- Temporary Flight Restrictions (TFRs) - 91.137-145:
 - These are where TFR NOTAMs derive authority.
- FAA fuel reserves - 91.151:
 - Day: 30 mins.
 - Night: 45 mins.
 - FIT: 60 mins day or night.
- Required documents and equipment - 91.205
 - This will be further explained in the airworthiness section.
- Aircraft lights - 91.209:
 - Used while the aircraft is being operated.
 - May be turned off when they are unsafe.
- Inoperative equipment - 91.213:
 - This will be further explained in the airworthiness section.

Flight Operations Manual

- Quiz the student on:
 - Student pilot limitations.
 - Weather minimums.
 - Wind velocities.
 - Required materials to bring for a stage check.
 - Drugs and alcohol.
 - Sterile cockpit policy.
 - Fuel reserves.

Certificates and Documents

- Medical certificates - 61.23.
 - Quiz the student on medical durations and privileges.
- Student pilot certification - 61.83.
 - Eligibility requirements - 61.83.
 - Application - 61.85.
 - What are the required endorsements for solo flight?
 - Pre-solo knowledge test.
 - Pre-solo flight training.
 - Solo flight (initial 90 days).
- Require logbook entries - 61.51.

Airworthiness

- What does airworthiness mean?
 - Type Certificate Data Sheet.
 - Ensure the student knows where to find it.
 - Faa.gov/aircraft → technical information → TCDS → search your aircraft.
 - What are the required inspections (AVIATES)?
 - Annual.
 - VOR.
 - 100 hours.
 - Altimeter.
 - Transponder.
 - ELT.
 - Static system.
 - Airworthiness directives:
 - What is an airworthiness directive?
 - Three types:
 - Notice of proposed rulemaking.
 - Final rule.
 - Emergency.
 - Service bulletins:
 - What is a service bulletin and how is it different from an AD?
- Required documents and equipment for VFR flight.
 - Applicable regulations: 91.203 and 91.205.
 - Required documents (ARROWES):
 - Airworthiness certificate.
 - Does not expire but invalid if not airworthy.
 - Check tail number and signature during pre-flight.
 - Registration certificate.
 - Typically expires every 7 years.
 - Check tail number and expiration date during pre-flight.
 - Different ways to lose the registration certificate:
 - Death.
 - Transfer.
 - Aircraft is found doing illegal activities.
 - Aircraft is totaled.
 - Certificate holder requests in writing to cancel their registration.
 - Certificate holder loses US Citizenship.
 - Radio station license / radio operator permit.
 - Obtained through the Federal Communications Commission (FCC).
 - Radio station license is for the airplane.

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- Expires every 10 years.
 - Radio operator permit is for the pilot.
 - Does not expire.
- Operational limitations.
 - POH section 02.
- Weight and balance.
 - POH section 06.
- Equipment list.
 - POH section 02.
 - Difference between MEL and KOEL.
- Supplements (FLICES):
 - POH section 09.
- Required equipment:
 - Day VFR required instruments (ATOMATOFLAMES) - 91.205.
 - Airspeed Indicator.
 - Tachometer.
 - For each engine.
 - Oil Pressure.
 - Manifold Pressure.
 - For each altitude engine (turbocharged/supercharged).
 - Altimeter.
 - Temperature Gauge.
 - For each liquid cooled engine.
 - Oil temperature.
 - Fuel Quantity Gauge.
 - Landing Gear Position Lights.
 - If retractable.
 - Anti-Collision Lights.
 - Aircraft certified after March 11th, 1996.
 - Magnetic direction indicator.
 - ELT.
 - Seatbelts.
 - Night VFR required instruments (FLAPS) - 91.205.
 - Fuses/resettable circuit breakers.
 - Landing Light (Electric) for hire.
 - Anti-Collision Lights.
 - Aircraft certified after August 11th, 1971.
 - Aircraft certified before that date STILL NEED anti-collision lights, but the standards/requirements are different.
 - Position Lights.

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- Source of Electricity (Alternator/Battery).
- Inoperative equipment - 91.213:
 - If an aircraft has a Minimum Equipment List, then refer to that first and find out if the aircraft is still airworthy.
 - Determining Airworthiness with inoperative equipment without MEL (TARK):
 - Type certificate data sheet (TCDS).
 - Airworthiness Directives.
 - Required for 91.205.
 - KOEL, Kinds of Operating Equipment List (section 2 of POH).
 - If the instrument or equipment is NOT required:
 - Remove from the airplane in accordance to 43.9 or deactivate and placard inoperative.

Airspace

- Controlled airspace (91.127 - 91.135).
 - General term that refers to the different classifications of airspace within which air traffic control (ATC) service is provided. (STAGE 1 WILL ONLY FOCUS ON THE FOLLOWING):
 - Class D:
 - Generally, from SFC to 2,500 ft MSL.
 - Surrounds airports that have a control tower.
 - Configuration can be different depending on the airport (KMLB!).
 - Usually designed to contain instrument approaches.
 - Entry requirements:
 - Establish two-way radio communication (ATC MUST say your call-sign).
 - Required equipment:
 - Two-way radios.
 - Class E:
 - Controlled airspace that is not class A, B, C, or D.
 - Extends up from either the surface or a designated altitude to the overlying or adjacent controlled airspace.
 - Sectional and other charts depict all locations of Class E airspace with bases below 14,500 feet MSL. In areas where charts do not depict a class E base, class E begins at 14,500 feet MSL.
 - In most areas, the Class E airspace base is 1,200 feet AGL. In many other areas, the Class E airspace base is either the surface or 700 feet AGL.
 - All airspace above FL 600 is Class E airspace.
 - Required equipment:
 - ADSB-OUT: at or above 10,000 feet msl, excluding airspace at and below 2,500 feet agl; over the Gulf of Mexico, at and above 3,000 feet msl, within 12 nm of the U.S. coast.

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- Mode C transponder: at and above 10,000 feet msl, excluding the airspace at and below 2,500 ft agl.
- Uncontrolled airspace (91.126)
 - Portion of the airspace system that is not designated as A, B, C, D, or E.
 - Class G:
 - Uncontrolled.
 - ATC has no authority or responsibility.
 - Although ATC has no authority, a pilot cannot be careless or reckless as this would break the 91.13 regulation.
 - VFR rules apply.
- VFR weather minimums:

Basic VFR Weather Minimums		
Airspace	Flight Visibility	Distance from Clouds
Class A	Not applicable	Not applicable
Class B	3 statute miles	Clear of clouds
Class C	3 statute miles	1,000 feet above 500 feet below 2,000 feet horizontal
Class D	3 statute miles	1,000 feet above 500 feet below 2,000 feet horizontal
Class E At or above 10,000 feet MSL Less than 10,000 feet MSL	5 statute miles	1,000 feet above 1,000 feet below 1 statute mile horizontal
	3 statute miles	1,000 feet above 500 feet below 2,000 feet horizontal
Class G 1,200 feet or less above the surface (regardless of MSL altitude). More than 1,200 feet above the surface but less than 10,000 feet MSL. More than 1,200 feet above the surface and at or above 10,000 feet MSL.	Day, except as provided in section 91.155(b) Night, except as provided in section 91.155(b)	1 statute mile 3 statute miles
	Day	1 statute mile
	Night	3 statute miles
		5 statute miles

Figure 15-8. Visual flight rule weather minimums.

- Special use airspace:
 - Where certain activities must be confined.
 - Limitations may be imposed on operations not participating.
 - Information for airspaces is located on aeronautical charts.
 - ONLY FOCUS ON THE FOLLOWING FOR STAGE 1:
 - Restricted area:
 - Area in which operations are hazardous to non-participating aircraft.
 - Artillery firing, guided missiles.
 - Flight of aircraft may be restricted, but not prohibited.
 - Must have permission from ATC or the controlling agency to enter.
 - Charted with an "R" followed by a number.
 - Information found on the back of charts.
 - Military Operation Area (MOA):
 - Defined vertical and lateral limits.
 - Separates certain military training activities from IFR traffic.

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- IFR traffic may be cleared through airspace if ATC can provide separation.
- Additional information for each found on the back of the sectional.
- Other airspace (ONLY FOCUS ON THE FOLLOWING):
 - Parachute jump aircraft operations:
 - Airports at which there are high numbers of parachute operations.
 - Published in chart supplement under Parachute Jumping Areas and depicted on sectional charts.
 - Temporary flight restrictions (TFRs):
 - Designated by flight data center (FDC) Notice to Airmen (NOTAM):
 - Begins with phrase “FLIGHT RESTRICTIONS” followed by location, effective time period, and defined in statute miles, and altitudes affected.
 - Purposes may include:
 - Protect persons or property in air or on surface from existing or imminent hazard.
 - Provide a safe environment for operation of disaster relief aircraft.
 - Prevent unsafe congestion of sightseeing aircraft above incident or event.
 - Protect declared national disasters for humanitarian reasons in Hawaii.
 - Protect the President, Vice President, or other public figures.
 - Provide a safe environment for space agency operations.
 - If penetrated, the pilot may undergo security investigations and certificate suspensions.

Radio Communications Failure

- Give the student a scenario where they have a radio communication failure in the practice area.
- Make sure they attempt to troubleshoot.
- If they decide to proceed to KMLB, quiz them on how they would enter the airspace/pattern and what light gun signals they are looking for.

ATC light gun signals

- Quiz the student on light gun signals.

Aircraft Systems

- Quiz the student on aircraft systems:
 - Aircraft structure.
 - Primary and secondary flight controls.
 - Powerplant.
 - Oil system.

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- Landing gear system.
- Hydraulic system.
- Fuel system.
- Pitot-static system.
- Vacuum system.
- Electrical system.

TOLD Card

- Have the student take out the TOLD card that was to be prepared for this lesson.
- Quiz the student on the following:
 - Weight and balance.
 - Arm, datum, moment, taxi fuel burn, and trip fuel burn.
 - Effects of forward and aft CG.
 - Effects of weight on performance.
 - Pressure altitude and density altitude.
 - What do they mean?
 - Why is it important?
 - Performance charts.
 - Ask them to show you how they got the takeoff and landing distances.
 - Aircraft speeds.
 - This should have been quizzed at the beginning of the lesson.
 - Ensure that the student can explain all of the different V speeds.

Aircraft Servicing

- What is preventative maintenance?
- Who can conduct preventive maintenance?
- What are some examples of preventative maintenance? (FAR Part 43).

Pre-Solo Knowledge Test

- Administer the pre-solo knowledge test.
- Must be corrected to 100%.
- Endorse the student and place the knowledge test in the student's folder.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 119.
- Assign homework:
 - Prepare a TOLD card to be used for the 119 lesson. (Continued on next page).

Completion Standards

The student should:

1. Demonstrate through the pre-solo knowledge test and oral briefing, the aeronautical knowledge necessary for solo flight operations, including regulations, limitations, aircraft procedures and systems.
2. Achieve a minimum score of 70% (corrected to 100%) on the pre-solo knowledge test prior to lesson completion.

Note: The instructor shall place the proper endorsement in the student's logbook [14 CFR Part 61.87(b)].